



Circuit Theory Course Project

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Part I

Sharif Analog Filter (SAF) Analysis

1 Introduction

Second-order filters are foundational building blocks in analog signal processing, widely used in audio systems, communication circuits, sensor interfaces, and instrumentation. Their behavior is described by the standard second-order transfer function:

$$H(s) = \frac{N(s)}{s^2 + \frac{\omega_n}{Q}s + \omega_n^2}, \quad (1)$$

where:

- ω_n is the natural frequency, setting the transition or resonance location,
- Q is the quality factor, determining damping or sharpness of the response,
- $N(s)$ determines whether the filter is low-pass, high-pass, or band-pass.

Only brief conceptual definitions are provided; students are required to research deeper mathematical interpretations independently.

2 Standard Second-Order Filters

All second-order filters share the denominator:

$$D(s) = s^2 + \frac{\omega_n}{Q}s + \omega_n^2.$$

2.1 Low-Pass Filter (LP)

$$H_{LP}(s) = \frac{A_{LP} \omega_n^2}{s^2 + \frac{\omega_n}{Q}s + \omega_n^2}$$

Cutoff frequency:

$$f_c = \frac{\omega_n}{2\pi}$$

2.2 High-Pass Filter (HP)

$$H_{HP}(s) = \frac{A_{HP} s^2}{s^2 + \frac{\omega_n}{Q}s + \omega_n^2}$$

Cutoff frequency:

$$f_c = \frac{\omega_n}{2\pi}$$

2.3 Band-Pass Filter (BP)

$$H_{BP}(s) = \frac{A_{BP}(\frac{\omega_n}{Q})s}{s^2 + \frac{\omega_n}{Q}s + \omega_n^2}$$

Center frequency:

$$f_c = \frac{\omega_n}{2\pi}$$

Bandwidth:

$$BW = f_2 - f_1$$

Relation to Q :

$$Q = \frac{f_c}{BW}.$$

3 The Sharif Analog Filter (SAF)

The Sharif Analog Filter (SAF) is a second-order, fully configurable active filter architecture consisting of:

- three operational amplifiers (A_1, A_2, A_3),
- two precision capacitors (C_1, C_2),
- internal resistors (R_1, R_2, R_4),
- external tuning resistors (R_G, R_Q, R_{F1}, R_{F2}).

Depending on the selected output node, the SAF can function as a low-pass filter, band-pass filter or high-pass filter. External resistors R_Q and R_G allow precise adjustment of the natural frequency ω_n and quality factor Q , enabling accurate control of bandwidth and resonance. The SAF combines internal precision components with user-defined tuning, achieving stable and repeatable performance. Students must:

- derive all transfer functions using Kirchhoff's laws and Laplace transforms,
- compute $H_{HP}(s), H_{BP}(s), H_{LP}(s)$,
- express all results in standard second-order form,
- in section 5 based on the desired target calculate R_{F1}, R_{F2}, R_G, R_Q
- now with calculated parameters, implement your circuit in LTspice then follow the requirements of section 6,
- based on section 7, compare your simulation's result with your theoretical exceptions

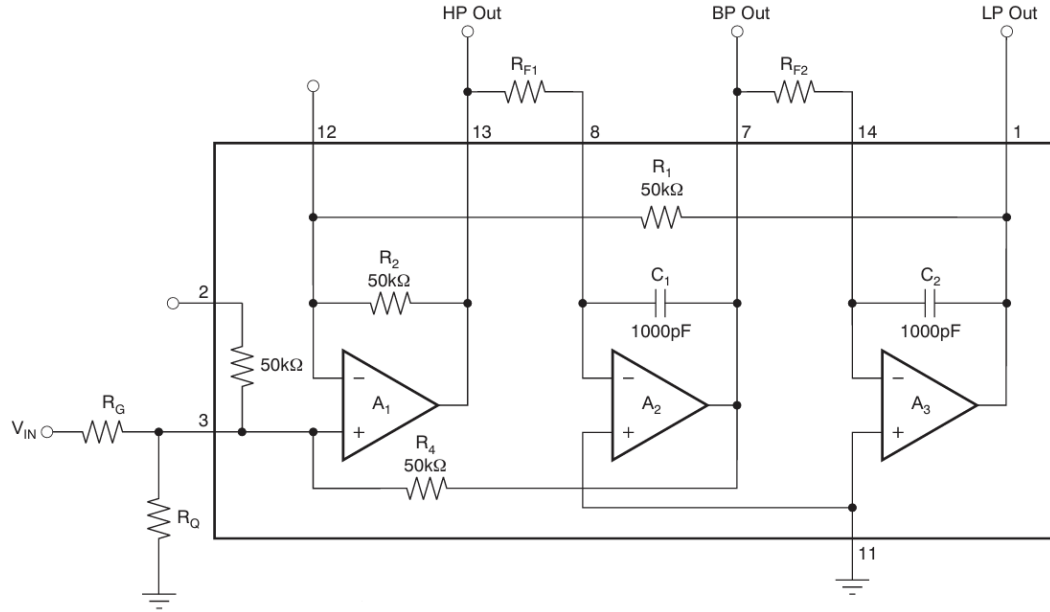


Figure 1: SAF Circuit Schematic

4 Transfer Functions

Using internal node expressions:

$$H_{HP}(s) = \frac{V_1}{V_i}, \quad H_{BP}(s) = \frac{V_2}{V_i}, \quad H_{LP}(s) = \frac{V_3}{V_i}$$

Rewrite each in standard form:

$$H(s) = \frac{N(s)}{s^2 + \frac{\omega_n}{Q}s + \omega_n^2}.$$

5 Filter Design

Find Q and f_c formulas in parameters based on your previous calculations then design the values for R_G , R_{F1} , R_{F2} , R_Q to achieve:

$$f_c = 20 \text{ kHz}, \quad Q = 0.75$$

Provide clear calculations

Assume the simplifying conditions:

$$R_G = R_Q, \quad R_{F1} = R_{F2}$$

6 Simulation Study

Using LTspice:

- draw filter configuration,

- run AC analysis from 1 kHz to 100 kHz,
- extract f_c , Q , and midband gain,
- include Bode magnitude plots.

7 Comparison and Discussion

Compare your analytical and simulated results. Summarize them in a table and calculate percentage differences in f_c and Q . Explain possible reasons for any observed differences, such as:

- Finite op-amp gain and bandwidth limitations
- Numerical tolerances in simulation
- Sensitivity of Q to resistor variations

Discuss how internal precision components contribute to performance stability, and reflect on why integrated active filter circuits are often preferred over discrete implementations.

8 Bonus: Notch Filter Design Using SAF

To create a notch (band-stop) filter using the SAF, recall the general second-order notch form:

$$H_{Notch}(s) = \frac{A_{Notch}(s^2 + \omega_n^2)}{s^2 + \frac{\omega_n}{Q}s + \omega_n^2}.$$

A notch can be constructed by combining:

- a low-pass term providing ω_n^2 in the numerator,
- a high-pass term providing s^2 in the numerator.

Thus:

$$H_{Notch}(s) = \alpha H_{LP}(s) + \beta H_{HP}(s)$$

for appropriate scaling.

- derive the notch transfer function directly from SAF node equations,
- express the result in standard form,
- simulate the SAF in notch configuration and verify the depth of attenuation.

9 Deliverables

Your final submission must include:

- A complete PDF report containing:
 - all mathematical derivations,
 - filter design calculations,
 - LTspice simulation plots,
 - comparison and discussion section,
 - bonus notch filter derivation and simulation.(if you did it)
- All LTspice schematic files (“.asc”).
- Any additional scripts or helper files used.

Part II

All-Pass Filters and Analog Noise Cancellation

10 All-pass filters

10.1 Introduction to all-pass filters

An all-pass filter is a specialized electronic circuit with a unique and seemingly contradictory frequency response: it allows all frequency components of an input signal to pass through to the output with no change in amplitude. In an ideal scenario, unlike low-pass or high-pass filters which attenuate specific frequency bands, an all-pass filter has a perfectly flat magnitude response across the entire frequency spectrum. Its defining function is not to alter the "loudness" or energy of the signal, but to systematically manipulate its timing, or phase. Specifically, it introduces a frequency-dependent phase shift, meaning that different frequency components within the signal are delayed by different amounts. This controlled phase manipulation is the primary reason all-pass filters are so valuable in electrical engineering. They are predominantly used for phase correction or equalization. In many practical systems, other components (like filters or amplifiers) can introduce unwanted, non-linear phase distortions that smear signals in time and can be particularly detrimental in data communication and audio processing. An all-pass filter can be carefully designed to apply an equal but opposite phase shift, thereby compensating for this distortion and restoring the original wave shape and timing relationships between frequencies.

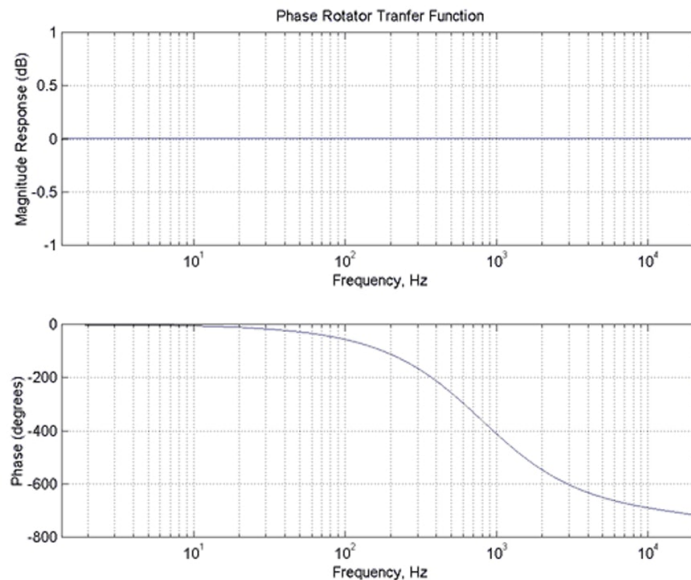


Figure 2: The magnitude and phase responses of an ideal all-pass filter.

10.2 Analysis of Analog Active All-Pass Filters

In this section, you will explore the design and properties of analog active all-pass filters. These circuits are unique because they alter the phase of a signal without affecting its amplitude. You will analyze their transfer functions and impulse responses to understand how they achieve this characteristic phase-shifting effect.

Part 1: First-Order All-Pass Filter

The circuit for a first-order active all-pass filter is shown in Figure 3.

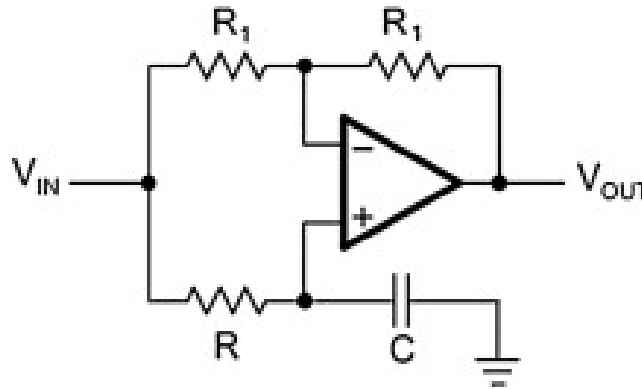


Figure 3: First-order active all-pass filter circuit

Tasks:

1. **Transfer Function:** Derive the transfer function, $H(s) = V_{\text{out}}(s)/V_{\text{in}}(s)$, for the given circuit.
2. **Frequency Response Analysis:**
 - From the transfer function, determine the magnitude response, $|H(j\omega)|$. Confirm that this is an “all-pass” filter.
 - Derive the phase response, $\angle H(j\omega)$. Explain how this phase shift affects an input signal.
 - Define the relationship between the phase shift value and the circuit elements (e.g., the resistor and capacitor values). What circuit parameter controls the frequency at which a specific phase shift occurs?

Part 2: Second-Order All-Pass Filter

The design can be extended to higher orders for a more complex phase response. The circuit for a second-order active all-pass filter is shown in Figure 4.

Tasks:

1. **Transfer Function:** Find the transfer function, $H(s)$, for this second-order circuit.
2. **Frequency Response Analysis:**
 - Determine the magnitude and phase response.
 - Explain how the phase shift effect differs from the first-order filter. How do the additional circuit elements provide more control over the phase response?
 - Define how the phase shift is determined by the values of the circuit components.

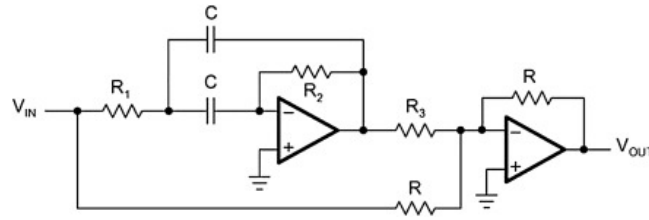


Figure 4: Second-order active all-pass filter circuit

Part 3: Investigating Higher-Order Filters

The pattern of adding stages can be continued to create all-pass filters of any order. Figure 5 conceptually represents a seventh-order all-pass filter constructed from multiple first and/or second-order sections.

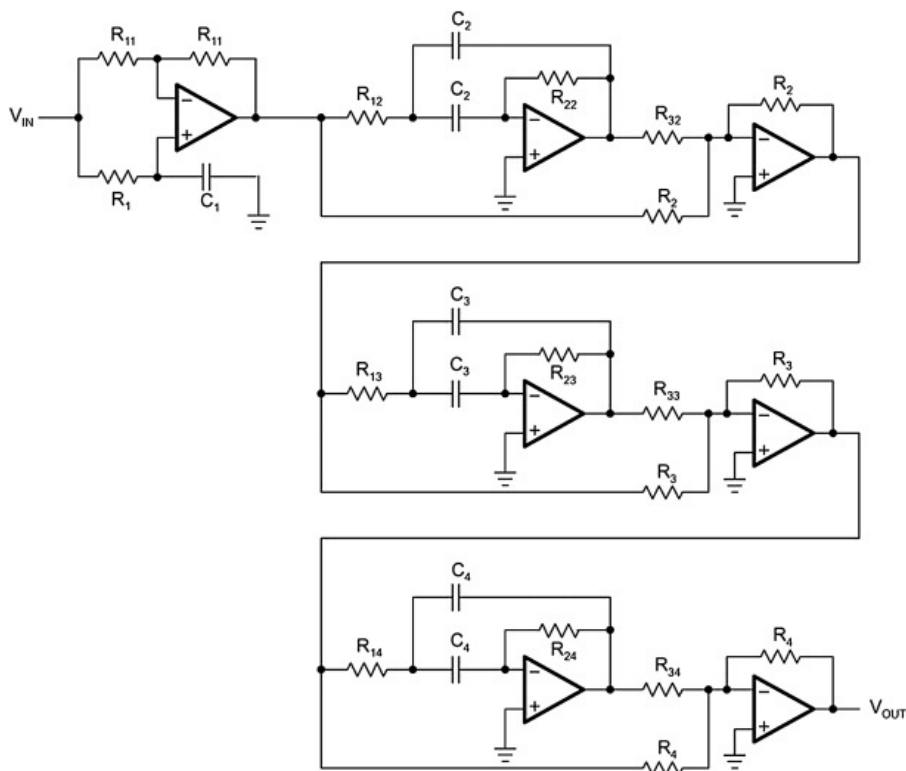


Figure 5: Block diagram of a seventh-order all-pass filter using cascaded stages

Question: Based on the patterns you observed in the first and second-order filters, predict how the **impulse response** of the filter changes as the order increases. Consider the following in your reasoning:

- What is the relationship between the order of a filter and the duration of its impulse response?
- How does increasing the number of stages affect the total phase shift a signal experiences?
- What would be the visual difference between the impulse response of a 1st-order and a 10th-order all-pass filter?

10.3 Design of an Analog Noise Cancelling Circuit Using an All-Pass Filter

In this section, we will design an active analog noise-cancelling (ANC) circuit. The fundamental principle behind ANC is to capture environmental noise and reproduce it with inverted phase (negative sign), so that when this anti-noise combines with the original noise, they destructively interfere and cancel each other out. This requires a microphone to capture the external noise, followed by signal processing that inverts and properly conditions the signal to create the cancelling waveform. To implement this system, we need to cascade three main components as shown in Figure 6:

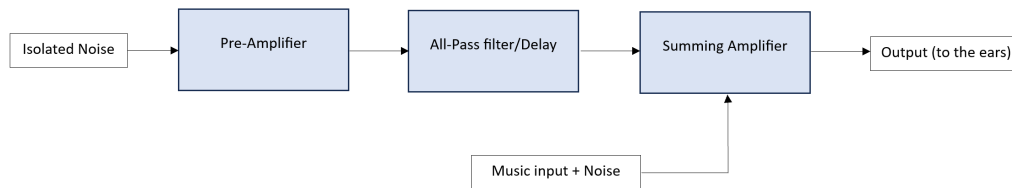


Figure 6: Block diagram of the analog noise-cancelling system

Question: In practical ANC systems, why is a delay component necessary between noise capture and cancellation?

Design Specifications

Design the following three stages using ideal op-amps:

1. **First Stage:** An amplifier with a gain of 10 to boost the captured noise signal
2. **Second Stage:** A circuit that introduces a precise 45 microsecond delay in the signal path
3. **Third Stage:** A subtraction circuit that combines the processed anti-noise with the original audio (signal + noise)

You will be provided with two audio inputs: the isolated noise signal and the combined music+noise signal.

Simulation Requirements

- Simulate your individual stage designs in LT-Spice and verify each stage functions correctly
- Test each stage using a simple sine wave input to validate performance
- **Primary deliverable:** Provide a detailed report for the second stage (delay circuit) simulation in LT-Spice, including:
 - Input and output waveforms demonstrating the 45 micro second delay
 - Frequency response analysis
 - Verification of the delay time across different test frequencies

10.4 System Integration: Cascading Stages for Complete ANC Implementation

In this final phase, we will integrate all previously designed stages to construct the complete active noise cancellation system. This system-level simulation will be performed in Simulink to evaluate the overall ANC performance. A comprehensive Simulink tutorial has been provided in the course materials. After reviewing the tutorial, construct your complete ANC system by connecting all stages in the proper sequence.

System Configuration

- **Input 1:** Reference noise signal (captured by external microphone)
- **Input 2:** Combined audio signal (music + ambient noise)
- **Output:** Noise-cancelled audio signal

Final Simulation Task

1. Build the complete cascaded system in Simulink using your previously designed stages
2. Apply the provided noise signal to the ANC processing chain
3. Feed the combined (music + noise) signal to the subtraction stage
4. Analyze the system output to evaluate noise cancellation performance

Deliverable Requirements

- Submit your complete Simulink model file
- Include the final output audio file demonstrating successful noise cancellation
- Provide a brief summary of the observed noise reduction performance
- Package all files (Simulink model, output audio, summary) in a single zip file for submission

Part III

Analysis of High-Frequency Response in Transistor Amplifiers

11 Introduction

At low to moderate frequencies, small-signal models for transistors (like the hybrid- π model) are incredibly effective. They treat the device as a controlled source with input and output resistances. However, as the operating frequency of a circuit increases into the MHz and GHz range, these simple models fail because they omit crucial physical effects. The primary limitation at high frequencies comes from the **internal parasitic capacitances** within the transistor. Any two conductive regions separated by an insulator or a depletion region (like the gate and channel, or the base-collector junction) form a small capacitor. At high frequencies, the impedance of these capacitors ($Z_C = 1/(sC)$) becomes small. This low impedance creates paths that can shunt the signal to ground or, more critically, feed the output signal back to the input, leading to gain reduction and potential instability. This project, will analyze these high-frequency effects.

12 High-Frequency Transistor Models

To accurately predict circuit behavior at high frequencies, we must augment the standard small-signal models to include these parasitic capacitances.

12.1 The MOSFET High-Frequency Model

The physical structure of the MOSFET leads to several key internal capacitances:

- C_{gs} (**Gate-to-Source Capacitance**): Formed by the gate oxide capacitance over the channel, plus the overlap of the gate with the source region.
- C_{gd} (**Gate-to-Drain Capacitance**): Primarily the capacitance from the overlap of the gate with the drain region. This is often the most critical parasitic due to the Miller effect.
- C_{sb} (**Source-to-Body Capacitance**): The junction capacitance of the source-to-substrate (body) p-n junction.
- C_{db} (**Drain-to-Body Capacitance**): The junction capacitance of the drain-to-substrate (body) p-n junction.

For most analyses, we assume the source is connected to the body (substrate), which shorts C_{sb} to ground. The C_{db} capacitance is often small and can be neglected for a simplified model. This results in the widely used high-frequency model shown in Figure 7.

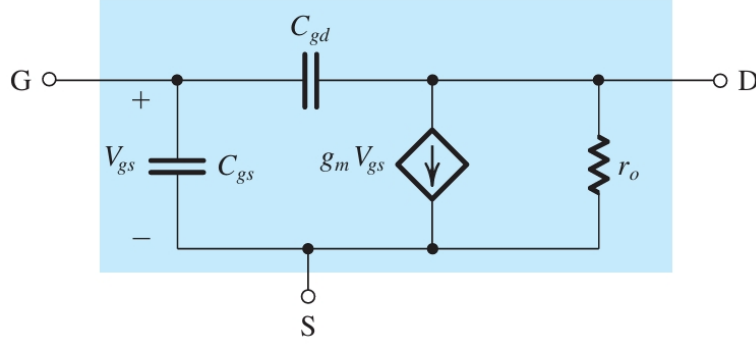


Figure 7: The simplified high-frequency small-signal model for a MOSFET, with the source connected to the body and C_{db} neglected.

12.1.1 Parametric Calculation (MOSFET)

The values for this model are derived from the device's physical construction and its DC operating point (bias). The key parametric equations are:

Transconductance (g_m):

$$g_m = \mu_n C_{ox} \frac{W}{L} |V_{OV}| = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} = \frac{2I_D}{|V_{OV}|} \quad (2)$$

Output Resistance (r_o):

$$r_o = \frac{|V_A|}{I_D} \quad (3)$$

where V_A is the Early voltage.

Gate-Source Capacitance (C_{gs}):

$$C_{gs} = \frac{2}{3} W L C_{ox} + W L_{ov} C_{ox} \quad (4)$$

This includes the capacitance of the channel ($\frac{2}{3} W L C_{ox}$) and the gate-source overlap capacitance ($W L_{ov} C_{ox}$).

Gate-Drain Capacitance (C_{gd}):

$$C_{gd} = W L_{ov} C_{ox} \quad (5)$$

This capacitance is primarily due to the gate-drain overlap (L_{ov}).

12.2 The BJT High-Frequency Model

The standard BJT hybrid- π model is augmented with two critical parasitic capacitances: C_π and C_μ .

- **C_π (Base-Emitter Capacitance):** This is the total capacitance seen between the base and emitter. It consists of two parts: $C_\pi = C_{de} + C_{je}$.
 - **C_{de} (Diffusion Capacitance):** This is the dominant component. It represents the charge (Q_n) that must be stored in the base region to sustain the collector current. It is proportional to I_C : $C_{de} = \tau_F g_m$, where τ_F is the forward base-transit time.

- C_{je} (**Base-Emitter Junction Capacitance**): This is the standard depletion capacitance of the forward-biased base-emitter p-n junction.
- C_{μ} (**Base-Collector Capacitance**): This is the depletion capacitance of the *reverse-biased* collector-base junction. Its value depends on the collector-base voltage (V_{CB}). Like C_{gd} , this is a feedback capacitor that is amplified by the Miller effect.

The resulting high-frequency model is shown in Figure 8. The model also includes the physical base spreading resistance, r_x .

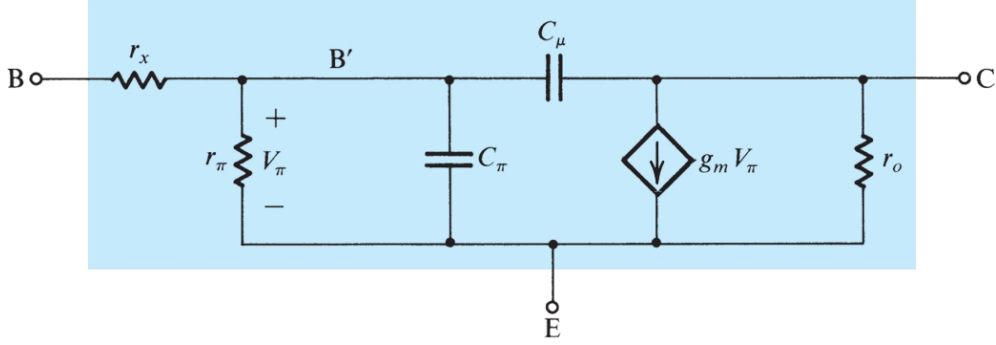


Figure 8: The high-frequency hybrid- π model for the BJT, including r_x , C_{π} , and C_{μ} .

12.2.1 Parametric Calculation (BJT)

The parameters for the BJT model are defined as follows:

Transconductance (g_m):

$$g_m = \frac{I_C}{V_T} \quad (6)$$

where V_T is the thermal voltage (approx. 25 mV at room temperature). **Input Resistance (r_{π}):**

$$r_{\pi} = \frac{\beta_0}{g_m} \quad (7)$$

where β_0 is the low-frequency DC current gain. **Output Resistance (r_o):**

$$r_o = \frac{|V_A|}{I_C} \quad (8)$$

where V_A is the Early voltage. **Total Base-Emitter Capacitance (C_{π}):**

$$C_{\pi} = C_{de} + C_{je} = (\tau_F g_m) + C_{je} \quad (9)$$

C_{je} is often approximated as $2C_{je0}$ (where C_{je0} is the zero-bias BE capacitance). **Base-Collector Capacitance (C_{μ}):**

$$C_{\mu} = \frac{C_{\mu 0}}{(1 + \frac{V_{CB}}{V_{0c}})^m} \quad (10)$$

where $C_{\mu 0}$ is the zero-bias capacitance, V_{0c} is the built-in voltage (e.g., 0.75 V), V_{CB} is the reverse-bias voltage, and m is the grading coefficient (e.g., 0.3 – 0.5).

13 Calculation Exercise: Model Parameters

Using the formulas presented in Section 12, calculate the complete set of high-frequency small-signal model parameters for the following two transistors.

13.1 Task 1: MOSFET Parameter Calculation

A typical n-channel MOSFET is fabricated in a process with the following characteristics and is biased at the given operating point.

- Process: $\mu_n C_{ox} = 400 \mu\text{A}/\text{V}^2$
- Geometry: $W = 20 \mu\text{m}$, $L = 0.5 \mu\text{m}$, $L_{ov} = 0.04 \mu\text{m}$
- Channel: $V_A = 20 \text{ V}$, $C_{ox} = 8.0 \text{ fF}/\mu\text{m}^2$
- DC Bias: $I_D = 500 \mu\text{A}$, $|V_{OV}| = 0.25 \text{ V}$

Required Calculations:

- The transconductance, g_m (using Eq. 2).
- The output resistance, r_o (using Eq. 3).
- The gate-source capacitance, C_{gs} (using Eq. 4).
- The gate-drain capacitance, C_{gd} (using Eq. 5).

13.2 Task 2: BJT Parameter Calculation

A typical NPN BJT is biased in the active region with the following parameters. Assume a room temperature of $V_T = 25 \text{ mV}$.

- Device: $\beta_0 = 120$, $V_A = 75 \text{ V}$, $\tau_F = 15 \text{ ps}$
- Capacitances: $C_{je0} = 2.0 \text{ pF}$, $C_{\mu 0} = 0.3 \text{ pF}$, $V_{0c} = 0.75 \text{ V}$, $m = 0.33$
- DC Bias: $I_C = 2.0 \text{ mA}$, $V_{CB} = 10 \text{ V}$

Required Calculations:

- The transconductance, g_m (using Eq. 6).
- The input resistance, r_π (using Eq. 7).
- The output resistance, r_o (using Eq. 8).
- The total base-emitter capacitance, C_π (using Eq. 9, assume $C_{je} \approx 2C_{je0}$).
- The base-collector capacitance, C_μ (using Eq. 10).

14 The Unity-Gain Frequency (f_T)

The parameters calculated in the previous section allow us to determine the ultimate high-frequency performance limit of the transistor itself. This limit is quantified by the **unity-gain frequency**, also known as the **transition frequency** (f_T). f_T is defined as the frequency at which the **short-circuit current gain** of the transistor drops to 1 (or 0 dB). It is a fundamental figure of merit that measures the "speed" of the device, independent of any external load resistors. To find it, we must derive the transfer function for the short-circuit current gain using Laplace analysis.

14.1 Task 3: Parametric Derivation of f_T

Your task is to derive the parametric equations for the unity-gain frequency (f_T) for both transistors using Laplace analysis.

14.1.1 Task 3.1: MOSFET f_T Derivation

Using the circuit configuration shown in Figure 9 (where the output is short-circuited), perform a Laplace analysis to find the transfer function for the short-circuit current gain, $A_i(s) = I_o(s)/I_i(s)$.

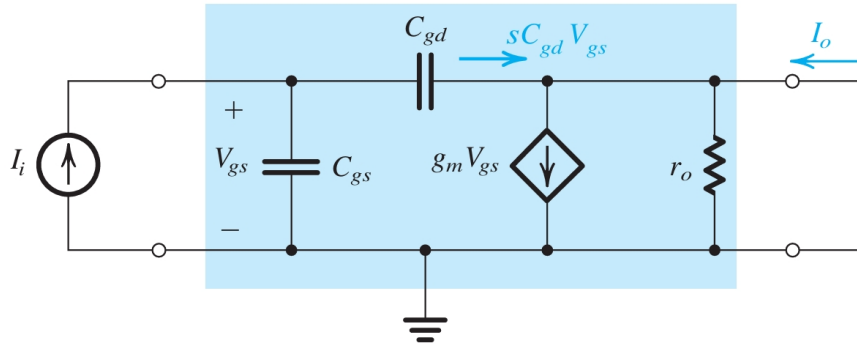


Figure 9: Circuit configuration for finding the short-circuit current gain $A_i(s)$ of the MOSFET.

Show all the intermediate steps to prove that the gain can be approximated as:

$$A_i(s) \approx \frac{g_m}{s(C_{gs} + C_{gd})} \quad (11)$$

From this, set $s = j\omega_T$ and solve for $|A_i(j\omega_T)| = 1$ to derive the final equation for f_T :

$$f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})} \quad (12)$$

(Hint: You may assume $g_m \gg |\omega C_{gd}|$ at high frequencies).

14.1.2 Task 3.2: BJT f_T Derivation

Similarly, use the circuit configuration in Figure 10 to find the short-circuit current gain $h_{fe}(s) = I_c(s)/I_b(s)$.

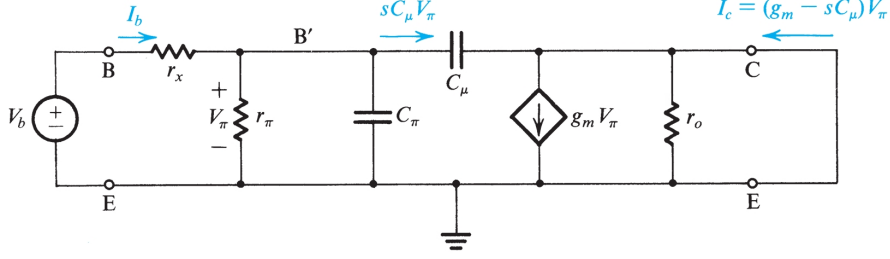


Figure 10: Circuit configuration for finding the short-circuit current gain $h_{fe}(s)$ of the BJT.

Show all the steps to find the transfer function $h_{fe}(s)$. Your derivation should lead to the standard single-pole approximation:

$$h_{fe}(s) \approx \frac{\beta_0}{1 + s/\omega_\beta} \quad (13)$$

Clearly define β_0 and the pole ω_β in terms of the model parameters (g_m, r_π, C_π, C_μ). Finally, show how this leads to the gain-bandwidth product relationship $\omega_T = \beta_0 \omega_\beta$ and derive the final equation for f_T :

$$f_T = \frac{g_m}{2\pi(C_\pi + C_\mu)} \quad (14)$$

14.2 Task 4: Numerical Calculation and Plotting

Based on the derivations in **Task 3** and the numerical values you calculated in Section 13, complete the following tasks.

14.2.1 Task 4.1: Calculate f_T and f_β

- **MOSFET:** Using your calculated values for g_m , C_{gs} , and C_{gd} from Task 1, find the numerical value for the unity-gain frequency, f_T , using your derived Eq. 12.
- **BJT:** Using your calculated values for g_m , r_π , C_π , C_μ , and β_0 from Task 2:
 1. Calculate the current-gain -3dB frequency, $f_\beta = \omega_\beta/(2\pi)$.

$$f_\beta = \frac{1}{2\pi r_\pi(C_\pi + C_\mu)} \quad (15)$$

2. Calculate the unity-gain frequency, f_T , using your derived Eq. 14.
3. Verify that $f_T \approx \beta_0 f_\beta$.

14.2.2 Task 4.2: Plot the Short-Circuit Current Gain

Create two types of plots for the short-circuit current gain for both the MOSFET ($A_i(s)$) and the BJT ($h_{fe}(s)$), from 1 MHz to 100 GHz.

- **Plot A (Normal Scale):** Plot the gain magnitude $|A(j\omega)|$ (on a linear y-axis) vs. frequency (on a linear x-axis).
- **Plot B (Bode Plot):** Plot the gain magnitude in decibels ($20 \log_{10} |A(j\omega)|$) (on a linear y-axis) vs. frequency (on a **logarithmic** x-axis).

14.2.3 Task 4.3: Analyze the Plots

Analyze the plots you just created.

- **Plot Comparison:** Examine the differences between Plot A (linear-linear) and Plot B (Bode plot). Which plot is more useful for finding f_T and f_β ? Explain why the Bode plot is the standard for frequency response analysis.
- **On the BJT Bode Plot:** Label the low-frequency gain ($20 \log_{10} \beta_0$).
- **On the BJT Bode Plot:** Find and label the -3dB frequency, f_β . Show that the gain at this point is approximately 3 dB below the low-frequency gain.
- **On both Bode Plots:** Find and label the unity-gain frequency (0 dB point), f_T . Verify that this matches your calculation from Task 4.1.
- **On both Bode Plots:** Label the roll-off slope, showing that it is -20 dB/decade.

15 High-Frequency Analysis of CS and CE Amplifiers

In the previous sections, we analyzed the transistor in isolation (specifically, with its output short-circuited) to find its internal parameters and theoretical speed limit, f_T . Now, we will analyze a complete amplifier circuit, including its source resistance (R_{sig}) and load resistance (R_L). When a load is present, the analysis becomes more complex due to the "feedback" capacitor that bridges the input and output: C_{gd} in the MOSFET and C_μ in the BJT. This capacitor is amplified by the gain of the circuit, creating a much larger *effective* capacitance at the input. This phenomenon is known as the **Miller Effect**.

15.1 The Miller Effect and the High-Frequency Pole (f_H)

Let's analyze the generic amplifier circuit shown in Figure 11. The capacitor C_f (representing C_{gd} or C_μ) connects the input (node V_i) to the output (node V_o).

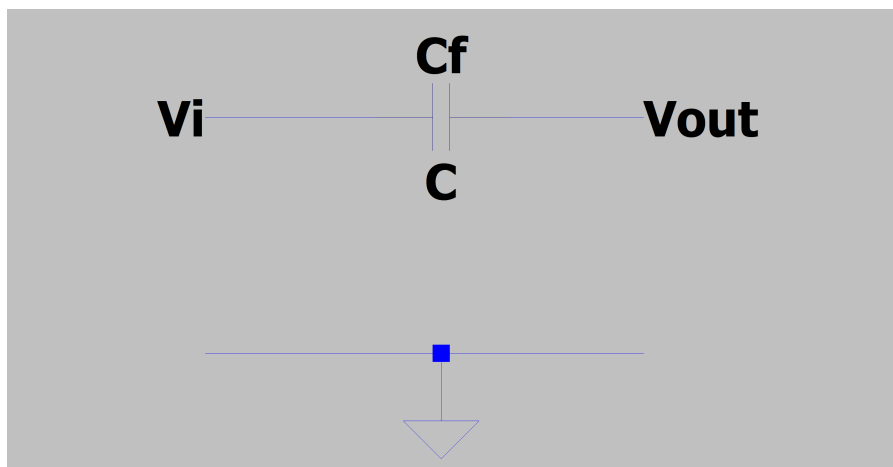


Figure 11: Conceptual circuit for Miller's Theorem.

The voltage gain from the input to the output is $A_v = V_o/V_i$. Using Laplace analysis, the current $I_f(s)$ flowing through C_f is:

$$I_f(s) = \frac{V_i(s) - V_o(s)}{1/(sC_f)} = sC_f(V_i(s) - V_o(s)) \quad (16)$$

Substituting $V_o(s) = A_v V_i(s)$:

$$I_f(s) = sC_f(V_i(s) - A_v V_i(s)) = sC_f(1 - A_v)V_i(s) \quad (17)$$

The input admittance $Y_{in}(s)$ seen by the input source due to this capacitor is $I_f(s)/V_i(s)$:

$$Y_{in}(s) = sC_f(1 - A_v) \quad (18)$$

This admittance is equivalent to an input capacitance, $C_{eq} = Y_{in}(s)/s$. For an inverting amplifier (like CS or CE), the mid-band gain A_v is a large negative number (e.g., $A_M = -50$).

$$C_{eq} = C_f(1 - A_M) \approx C_f(1 - (-|A_M|)) = C_f(1 + |A_M|) \quad (19)$$

This "Miller Capacitance" (C_{eq}) is the small feedback capacitor C_f multiplied by the magnitude of the amplifier's gain. This large capacitance appears in parallel with the transistor's own input capacitance (C_{gs} or C_π), creating a total input capacitance:

$$C_{in} = C_{\text{input_base}} + C_{eq} \quad (\text{e.g., } C_{in} = C_{gs} + C_{gd}(1 + |A_M|)) \quad (20)$$

This large C_{in} forms an RC low-pass filter with the equivalent resistance seen by the input, R'_{sig} . This filter creates a **dominant low-frequency pole**, ω_H , which determines the amplifier's -3dB bandwidth.

$$\omega_H = \frac{1}{R'_{sig} C_{in}} \quad (21)$$

The overall amplifier transfer function can therefore be approximated as a single-pole low-pass system:

$$A_v(s) = \frac{A_M}{1 + s/\omega_H} \quad (22)$$

where A_M is the mid-band (low-frequency) gain.

15.2 Common-Source (CS) Amplifier Analysis

The high-frequency model of a full CS amplifier is shown in Figure 12.

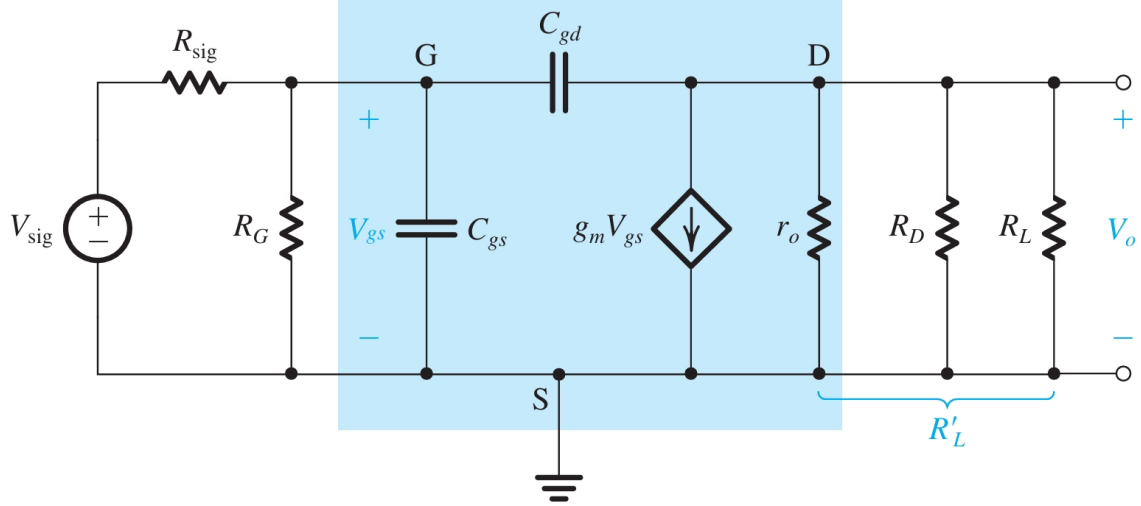


Figure 12: High-frequency equivalent circuit of the CS amplifier.

Based on the Miller analysis:

- **Mid-band Gain (A_M):** $A_M = \frac{V_o}{V_{sig}} = - \left(\frac{R_G}{R_G + R_{sig}} \right) (g_m R'_L)$
- **Load Resistance (R'_L):** $R'_L = r_o \parallel R_D \parallel R_L$
- **Input Resistance (R'_{sig}):** $R'_{sig} = R_{sig} \parallel R_G$
- **Total Input Capacitance (C_{in}):** $C_{in} = C_{gs} + C_{gd}(1 + g_m R'_L)$
- **High-Frequency Pole (ω_H):** $\omega_H = \frac{1}{R'_{sig} C_{in}}$

15.3 Common-Emitter (CE) Amplifier Analysis

The analysis for the CE amplifier (Figure 13) is exactly analogous.

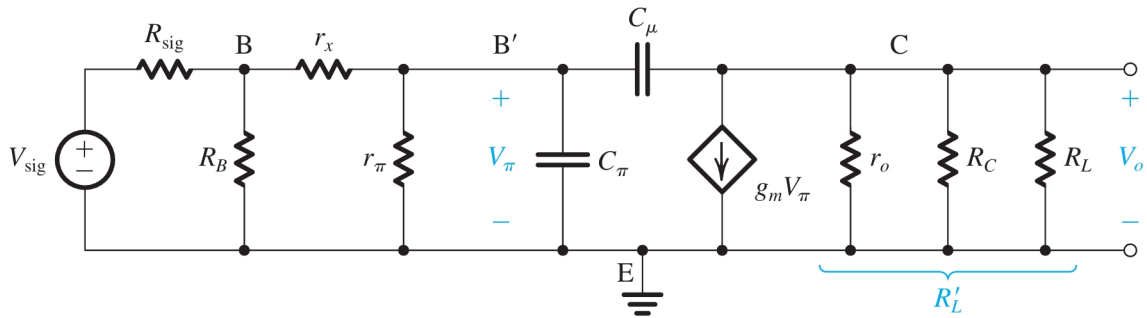


Figure 13: High-frequency equivalent circuit of the CE amplifier.

Based on the Miller analysis (and using the simplified input resistance from Fig 9c):

- **Mid-band Gain (A_M):** $A_M = \frac{V_o}{V_{sig}} \approx - \left(\frac{R_B \parallel (r_x + r_\pi)}{R_{sig} + (R_B \parallel (r_x + r_\pi))} \right) (g_m R'_L)$
- **Load Resistance (R'_L):** $R'_L = r_o \parallel R_C \parallel R_L$

- **Input Resistance (R'_{sig}):** $R'_{sig} = (R_{sig} \parallel R_B) \parallel (r_x + r_\pi)$ (Note: ECE 255 Fig 9b uses a more complex Thevenin model, but this is a common and close approximation).
- **Total Input Capacitance (C_{in}):** $C_{in} = C_\pi + C_\mu(1 + g_m R'_L)$
- **High-Frequency Pole (ω_H):** $\omega_H = \frac{1}{R'_{sig} C_{in}}$

15.4 Task 5: Amplifier Gain, Bandwidth, and Plotting

For this task, you will use the transistor parameters you calculated in Section 13 to analyze the performance of a complete amplifier.

15.4.1 Task 5.1: Parametric Transfer Function

Write out the full, parametric (symbolic) expression for the voltage gain $A_v(s) = V_o(s)/V_{sig}(s)$ for the **CS amplifier** shown in Figure 12. Your final answer should be in the standard form $A_v(s) = \frac{A_M}{1+s/\omega_H}$. Explicitly define A_M and ω_H in terms of the given components ($g_m, r_o, C_{gs}, C_{gd}, R_{sig}, R_G, R_D, R_L$).

15.4.2 Task 5.2: Numerical Calculation

Use the transistor parameters you calculated in **Task 1 (MOSFET)** and **Task 2 (BJT)** to calculate the amplifier performance metrics for the two circuits below. **Circuit 1: CS Amplifier**

- **Transistor:** Use your MOSFET parameters from Task 1.
- **External Components:** $R_{sig} = 50 \text{ k}\Omega$, $R_G = 10 \text{ M}\Omega$, $R_D = 10 \text{ k}\Omega$, $R_L = 10 \text{ k}\Omega$.
- **Calculate:** 1. The combined load resistance R'_L . 2. The mid-band voltage gain, A_M . 3. The total input capacitance, C_{in} (note the contribution of the Miller effect). 4. The combined input resistance R'_{sig} . 5. The upper 3dB frequency, $f_H = \omega_H/(2\pi)$.

Circuit 2: CE Amplifier

- **Transistor:** Use your BJT parameters from Task 2, and assume $r_x = 100 \text{ }\Omega$.
- **External Components:** $R_{sig} = 10 \text{ k}\Omega$, $R_B = 100 \text{ k}\Omega$ (biasing resistor), $R_C = 5 \text{ k}\Omega$, $R_L = 5 \text{ k}\Omega$.
- **Calculate:** 1. The combined load resistance R'_L . 2. The mid-band voltage gain, A_M . 3. The total input capacitance, C_{in} . 4. The combined input resistance R'_{sig} . 5. The upper 3dB frequency, $f_H = \omega_H/(2\pi)$.

15.4.3 Task 5.3: Plotting the Frequency Response

For both the CS and CE amplifiers, use your calculated numerical values for A_M and f_H to create a Bode plot of the voltage gain $A_v(s)$.

- The plot should show the gain magnitude in decibels (dB): $20 \log_{10} |A_v(j\omega)|$.
- The frequency axis should be logarithmic, from 100 Hz to 1 GHz.

15.4.4 Task 5.4: Analysis from the Plots

Analyze the two Bode plots you just created.

- Clearly label the mid-band gain (in dB) on each plot.
- Find and label the **-3dB frequency** (f_H) on each plot. Verify that the gain at this frequency is 3 dB lower than the mid-band gain and that it matches your calculation from Task 5.2.
- Label the roll-off slope and verify that it is -20 dB/decade after f_H .
- Find and label the **Unity-Gain Frequency** (where the gain line crosses 0 dB).
- Calculate the Gain-Bandwidth Product ($GBW = |A_M| \times f_H$) for each amplifier and compare it to the unity-gain frequency you found on the plot.

16 RF Input Impedance Matching

In the previous sections, we analyzed the voltage gain of standard CS and CE amplifiers. We found that their bandwidth is limited by a dominant pole ω_H , which is largely created by the Miller-multiplied capacitance C_{in} at the input. In RF (Radio Frequency) design, we have an additional challenge. Most RF systems (like antennas, filters, and other amplifiers) are designed to have an impedance of $50\ \Omega$. To transfer maximum power and prevent signal reflections, we must ensure that the input impedance (Z_{in}) of our amplifier is also $50\ \Omega$. The C_{in} we calculated earlier (e.g., $C_{in} = C_{gs} + C_{gd}(1 + |A_M|)$) makes the input impedance of a simple CS/CE stage highly capacitive:

$$Z_{in} \approx \frac{1}{sC_{in}} \quad (23)$$

This capacitive impedance creates a severe mismatch with a $50\ \Omega$ source. This section explores common techniques using inductors to "match" this impedance, forcing Z_{in} to become purely resistive (i.e., $Z_{in} = 50\ \Omega$) at a specific operating frequency, ω_0 .

16.1 Solution 1: CS/CE with Inductive Degeneration

One of the most powerful and common techniques is to add a small inductor in the source (or emitter) path, known as a "degeneration inductor" (L_s or L_e). This technique not only helps with stability but also fundamentally changes the input impedance. Let's analyze the input impedance Z_{in} of the CS stage in Figure 14. To simplify the core concept, we will (for this analysis only) ignore the Miller effect from C_{gd} and focus on the interaction between L_s and C_{gs} . This is a valid assumption in modern RF design, where the C_{gd} feedback is often removed by using a Cascode topology.

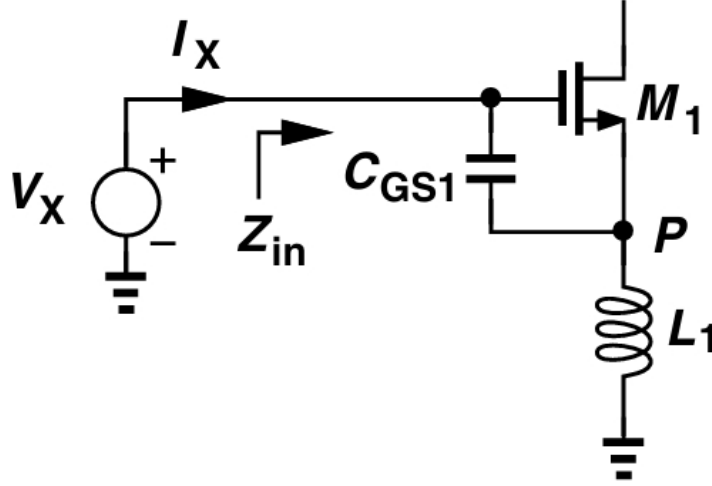


Figure 14: Circuit for analyzing input impedance with inductive degeneration.

The input current $I_x(s)$ flows only through C_{gs} , so:

$$I_x(s) = sC_{gs} \cdot V_{gs}(s) \implies V_{gs}(s) = \frac{I_x(s)}{sC_{gs}} \quad (24)$$

The input voltage $V_x(s)$ is the sum of $V_{gs}(s)$ and the voltage at the source, $V_s(s)$:

$$V_x(s) = V_{gs}(s) + V_s(s) \quad (25)$$

Assuming the drain current $I_d(s) = g_m V_{gs}(s)$ is the only current flowing through L_s :

$$V_s(s) = I_d(s) \cdot (sL_s) = (g_m V_{gs}(s)) \cdot (sL_s) = sg_m L_s V_{gs}(s) \quad (26)$$

Substituting $V_s(s)$ back into the equation for $V_x(s)$:

$$V_x(s) = V_{gs}(s) + sg_m L_s V_{gs}(s) = V_{gs}(s)(1 + sg_m L_s) \quad (27)$$

Now, by substituting the expression for $V_{gs}(s)$ from the first equation, we can find $Z_{in}(s) = V_x(s)/I_x(s)$:

$$Z_{in}(s) = \frac{V_x(s)}{I_x(s)} = \frac{V_{gs}(s)(1 + sg_m L_s)}{sC_{gs} V_{gs}(s)} = \frac{1 + sg_m L_s}{sC_{gs}} \quad (28)$$

Separating the real and imaginary parts gives the profound result:

$$Z_{in}(s) = \frac{g_m L_s}{C_{gs}} + \frac{1}{sC_{gs}} \quad (29)$$

This derivation shows that adding a source inductor L_s magically creates a **real resistance** ($R_{in} = g_m L_s / C_{gs}$) in series with the original gate-source capacitance.

16.2 Solution 2: CS with Inductive (Tuned) Load

Another common RF technique is to replace the drain resistor (R_D) with an inductor, L_D . This inductor forms a parallel resonant "tank" circuit with the output capacitance of the amplifier (C_{out}). This is called a **tuned amplifier**.

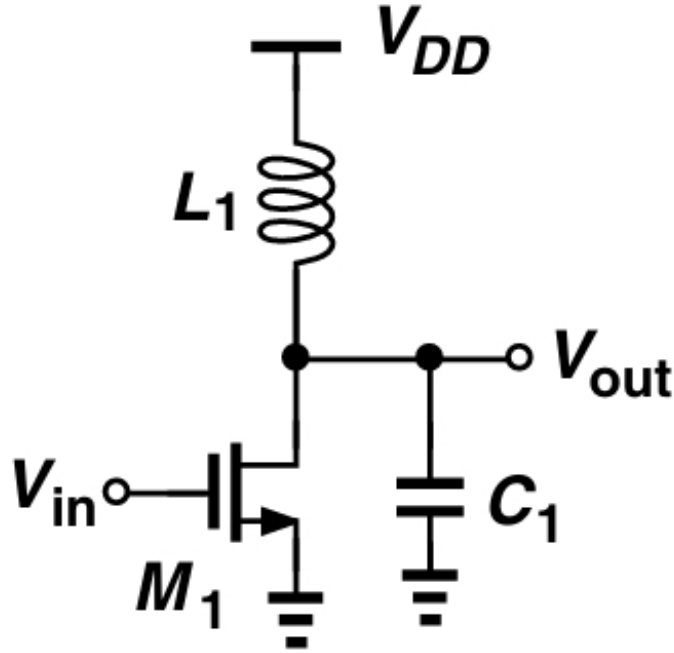


Figure 15: CS Amplifier with an Inductive (Tuned) Load.

This technique primarily shapes the *output* impedance and provides high gain only at the resonant frequency, $\omega_0 \approx 1/\sqrt{L_D C_{out}}$. However, it also has a strong effect on the *input* impedance.

16.3 Task 6: Achieving a 50 Ω Input Match

Our Z_{in} from Eq. 29 is still complex. To make it purely resistive and equal to 50 Ω , we must do two things:

1. Add a series "matching" inductor, L_g , at the gate to cancel the capacitive part.
2. Choose L_s to set the real part to 50 Ω .

The total input impedance $Z_{match}(s)$ becomes:

$$Z_{match}(s) = sL_g + Z_{in}(s) = sL_g + \frac{g_m L_s}{C_{gs}} + \frac{1}{sC_{gs}} \quad (30)$$

16.3.1 Task 6.1: Parametric Derivation for L_s and L_g

You are designing an RF amplifier that must have an input impedance of $Z_{in} = 50 \Omega$ at an operating frequency ω_0 .

- **1. Find L_s :** To make the real part of the impedance 50 Ω , show the equation for L_s .
- **2. Find L_g :** To make the input impedance purely resistive, the imaginary parts must cancel at ω_0 . Set the imaginary part of $Z_{match}(j\omega_0)$ to zero and derive the equation for the required L_g .

16.3.2 Task 6.2: Qualitative Discussion (Inductive Load)

- Using your knowledge of the Miller Effect (Section 15), qualitatively explain how replacing R_D with an inductor L_D (as in Figure 15) would affect the input impedance Z_{in} of the CS amplifier.
- Which specific component from the high-frequency model (Figure 7) links the input and output, causing this effect?
- The gain $A_v(s)$ of a tuned amplifier is complex and changes rapidly around ω_0 . What does this imply about the Miller capacitance $C_{eq} = C_{gd}(1 - A_v(s))$? (Hint: Can C_{eq} become negative?)

17 Tuned Amplifiers with Transformer Loads

In our RF matching analysis (Section 16), we used inductors to create a purely resistive input impedance. Another critical application of inductors, particularly transformers, is to create a **tuned load**. Instead of a simple resistor R_D , we can use the primary winding of a transformer as the load for the CS (or CE) amplifier. The key insight is that the transformer's primary winding has an inductance, L_p . This inductance will **resonate** with the total output capacitance of the circuit, C_{out} (which includes the transistor's C_{db} , C_{gd} , and any stray capacitance). This creates a parallel **RLC tank circuit** as the load. This circuit has a very high impedance at one specific frequency (the resonant frequency, ω_0) and a low impedance at all other frequencies. The result is an amplifier with "tunable" gain: it provides very high gain, but only in a narrow band around ω_0 . This is the fundamental principle of most RF amplifiers.

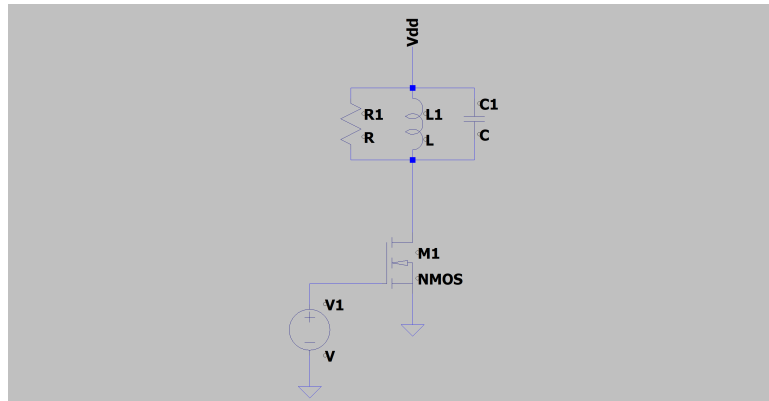


Figure 16: CS Amplifier with a Transformer-Coupled Tuned Load.

17.1 Task 7: Analysis of a Tuned Load

Consider the circuit in Figure 16. The transistor (with parameters g_m , r_o , and C_{out}) drives a transformer load. The transformer has a primary inductance L_p and reflects the final load R_L to the primary side as a parallel resistance $R_p = R_L/n^2$, where n is the turns ratio. The total load impedance $Z_{load}(s)$ seen by the dependent current source $g_m V_{gs}$ is a parallel RLC circuit:

$$Z_{load}(s) = r_o \parallel R_p \parallel (sL_p) \parallel \left(\frac{1}{sC_{out}} \right) \quad (31)$$

Let's define $R_{total} = r_o \parallel R_p$. The impedance is:

$$Z_{load}(s) = \frac{1}{\frac{1}{R_{total}} + sC_{out} + \frac{1}{sL_p}} = \frac{s/C_{out}}{s^2 + s\frac{1}{R_{total}C_{out}} + \frac{1}{L_pC_{out}}} \quad (32)$$

17.1.1 Task 7.1: Derive the Gain Function

Using Laplace analysis, show that the voltage gain $A_v(s) = V_o(s)/V_{in}(s)$ is given by:

$$A_v(s) = -g_m \cdot Z_{load}(s) = -g_m \left(\frac{s/C_{out}}{s^2 + s\frac{1}{R_{total}C_{out}} + \frac{1}{L_pC_{out}}} \right) \quad (33)$$

Explain why this is a **band-pass** filter response.

17.1.2 Task 7.2: Find Resonant Frequency and Peak Gain

- Find the resonant frequency ω_0 (the frequency of peak gain).
- Show that the gain at this resonant frequency, $A_v(j\omega_0)$, is purely real (no imaginary part) and is given by:

$$A_v(j\omega_0) = -g_m R_{total} = -g_m(r_o \parallel R_p) \quad (34)$$

17.1.3 Task 7.3: Find the Bandwidth

The quality factor, Q , of a parallel RLC circuit is $Q = R_{total}/(\omega_0 L_p) = R_{total}(\omega_0 C_{out})$. The bandwidth (BW) of the amplifier is given by $BW = \omega_0/Q$.

- Derive the expression for the amplifier's -3dB bandwidth, BW , in terms of R_{total} and C_{out} .
- What is the trade-off between getting a very high gain (large R_{total}) and achieving a wide bandwidth?

References

1. Sedra, A. S., & Smith, K. C. (2020). *Microelectronic Circuits*. Oxford University Press.
2. Razavi, B. (2012). *RF Microelectronics*. Pearson.